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Project Proposal

1. Project Description

   Luneth GlassVault is a glassware inventory and sales-logging app but for a real retail store ‘Luneth Glassware’ operating in the town of Puerto Galera, Oriental Mindoro. It is built to be used by multiple (at least 2) employees. The employees can look up incoming shipments by Factura (invoice), check existing inventory (supplier, price, item name, and code), and log customer sales. Every sale and inventory entry is partitioned by employee accounts using Realm.

2. RealmObject Subclasses

   The Realm will be composed of two main models

   A. FacturaItem.java - Represents the Inventory Stock
      a. `code` (`String`, `@PrimaryKey`) – Item/Product Barcode or Internal Code.
      b. `facturaNumber` (`String`) – Supplier invoice reference code.
      c. `supplier` (`String`) – Name of the glassware manufacturer/supplier.
      d. `name` (`String`) – Product name (e.g., *"Crystal Wine Glass 350ml"*).
      e. `price` (`double`) – Selling price per unit.
      f. `imageUri` (`String`) – File path to the photo snapped via **Camera**.
      g. `stockQuantity` (`int`) – Remaining quantity available.

   B. SaleLog/.java - Which handles logged transactions
      a. `uuid` (`String`, `@PrimaryKey`) – Unique transaction ID.
      b. `userUuid` (`String`) – **Multi-user partitioning field.** UUID of the employee who made the sale.
      c. `itemCode` (`String`) – Linked product code.
      d. `facturaNumber` (`String`) – Linked supplier invoice.
      e. `quantitySold` (`int`) – Units sold to customer.
      f. `totalPrice` (`double`) – Calculated total (`price * quantity`).
      g. `timestamp` (`long` or `String`) – Date/time of the sale.

3. Screen Breakdown

Login	EditTexts, Buttons	Employee logs in using Name and Password; saves logged-in userUuid to SharedPreferences
Admin / User Mgmt	RecyclerView	Admin view to edit or clear employee accounts
1 Quick Shift Hub	2 Big Buttons + 2 TextViews	This screen shows: 1) Welcome Employee Name, 2) Total Items Sold Today counter 3) Button to "Log New Sale",

		4) Button to "View Facturas".
2 Factura Catalog	RecyclerView, Adapter	Lists all registered glassware and facturas (Supplier, Code, Price, Thumbnail). Clicking an item opens Screen 4.
3 Log New Sale / Add Item	EditTexts, ImageView, Camera Button	Selects an item code/factura, snaps a photo of the product/receipt with the Camera, and saves the sale under the current employee's userUuid.
4 Factura & Item Details	Large ImageView, TextViews, Action Buttons	Displays full product photo, supplier name, code, price, and remaining stock. Has a direct button: "Sell This Item" (opens Screen 3 pre-filled).
5 My Sales History	RecyclerView	A simple vertical list displaying only the logged-in employee's sales using .equalTo("userUuid", currentUuid). Shows Item Name, Factura #, and Total Amount.
6 Employee Profile	ImageView, TextViews, Logout Button	Displays logged-in employee name, role, and a Logout button that clears SharedPreferences and redirects to the Login screen.